

It may be remarked that in the present case, assuming the invariance of the function in y , we may obtain the transformed equation in a very simple manner by writing in the first instance $C = 0$; this gives

$$(a, b, c, d \mid x, 1)^3 = 0, \quad y = (ax + b)B,$$

and thence

$$\frac{1}{a}(a, b, c, d \mid y - bB, aB)^3 = 0,$$

or developing,

$$y^3 + 3y(ac - b^2)B^2 + (a^2d - 3abc + 2b^3)B^3 = 0,$$

where the expressions for the coefficients are to be completed by the consideration that these coefficients are covariants of the form $(a, b, c, d \mid B, C)^3$. But it is only in the case in hand of a cubic equation that the transformed equation can be obtained in this manner.

In the case of a quartic equation, we have

$$(a, b, c, d, e \mid x, 1)^4 = 0,$$

$$y = (ax + b)B + (ax^2 + 2bx + 3c)C + (ax^3 + 3bx^2 + 6cx + 4d)D,$$

and these give the system of equations

$$(y - bB - 3cC - 4dD, -aB - 4bC - 6cD, -aC - 4bD, -aD \mid 1, x, x^2, x^3) = 0,$$

$$\begin{vmatrix} y - bB - 3cC - 4dD & -aB - 4bC - 6cD & -aC - 4bD & -aD \\ eD & y - bB - 3cC + dD & -aB - 4bC & -aC \\ eC & 4dC + eD & y - bB + 3cC + dD & -aB \\ eB & 4dB + eC & 6cB + 4dC + eD & y + 3bB + 3cC + dD \end{vmatrix} = 0,$$

and the transformed equation is therefore found by equating to zero the determinant formed out of the matrix contained in this equation.

The developed result, which was obtained by a different process in the “Deuxième Note” above referred to, is

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4 = 0,$$

where the coefficients $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}$ are quartic, sextic, and octic functions of the form $(a, b, c, d, e \mid B, C, D)^k$ in the two sets of variables; the detailed expressions, given in the original on this and the following pages as a two-dimensional array of monomial coefficients in (a, b, c, d, e) keyed against the monomial products of (B, C, D) , are reproduced below:

[Large multivariate coefficient table for $\frac{1}{4}\mathfrak{C}$, expressed as a sum of monomial products $\text{monom}(a, b, c, d, e) \cdot \text{monom}(B, C, D)$ with small integer coefficients (typical entries: $+3, -2, +6$); the table spans the lower half of printed page 379 of the original. The OCR of this dense table is not trustworthy and the table is here represented as a placeholder; the cleanly typeset expressions for the binary forms U', H', Φ' on page 380, together with the identities for $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}$ given there in terms of U', H', Θ' , suffice for all further use.]

[The remainder of the expression for $\frac{1}{4}\mathfrak{C}$ continues at the top of page 380 in the original, together with the analogous (much larger) multivariate coefficient tables for $\frac{1}{8}\mathfrak{D}$ and $\frac{1}{16}\mathfrak{E}$. These are large two-dimensional arrays of small integer coefficients indexed by the monomials of a, b, c, d, e and of B, C, D ; the OCR of these tables is not faithful. Cayley's purpose in displaying them is to enable the next identity (below), which expresses the same three quantities compactly in terms of invariants U', H', Φ', Θ' of the auxiliary quartic. We retain the prose, the new notation, and the identity, all of which are independently legible.]

I write

$$U' = aB^3 + 4bB^2C + c(2BD + 4C^2) + 4dCD + eD^2,$$

$H' = (ac - b^2)B^2 + 2(ad - bc)BC + (ae - 2bd + c^2)BD + 4(bd - c^2)C^2 + 2(be - cd)CD + (ce - d^2)D^2$,
and I represent by Φ' the expression which has just been found for $\mathfrak{E}$. These functions U', H', Φ' are invariants of the two forms

$$(a, b, c, d, e \mid X, Y)^4, \quad (B, C, D \mid Y, -X)^2;$$

we have, moreover, the invariants

$$I = ae - 4bd + 3c^2, \quad J = ace - ad^2 - b^2e + 2bcd - c^3$$

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which I represent as usual by I, J , and the invariant $BD - C^2$, which I represent by Θ' . This being so, we have

$$\begin{aligned} \mathfrak{E} &= 6H' - 2I\Theta', \\ \mathfrak{D} &= 4\Phi', \\ \mathfrak{E} &= IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H', \end{aligned}$$

the last of which may be verified as follows:—viz., writing $a = e = 1$, $b = d = 0$, $c = \theta$, it becomes

$$\begin{aligned} &(1 + 3\theta^2)[B^2 + \theta(2BD + 4C^2) + D^2]^2 \\ &\quad - 3[\theta B^2 + (1 + \theta^2)BD - 4\theta^2C^2 + \theta D^2]^2 \\ &\quad + (1 + 3\theta^2)^2(BD - C^2)^2 \\ &\quad + 12(\theta - \theta^3)(BD - C^2)[B^2 + \theta(2BD + 4C^2) + D^2] \\ &\quad + 2(1 + 3\theta^2)(BD - C^2)[\theta B^2 + (1 + \theta^2)BD - 4\theta^2C^2 + \theta D^2] \\ &= B^4 \\ &\quad + B^3D (12\theta) \\ &\quad + B^2C^2 (-6\theta + 54\theta^3) \\ &\quad + B^2D^2 (2 + 36\theta^2) \\ &\quad + BC^2D (-4 + 36\theta^2) \\ &\quad + C^4 (1 - 18\theta^2 + 81\theta^4) \\ &\quad + CD^2 (-6\theta + 54\theta^3) \\ &\quad + BD^3 (12\theta) \\ &\quad + D^4, \end{aligned}$$

which is an identical equation. [This is in effect the identity of the “Deuxième Note,” see ante p. 372, but the left-hand side was in the original printed incorrectly, with four lines only instead of five.]

The expression for the invariant I (quadrinvariant) of the function $(1, 0, \mathfrak{E}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$ is $\mathfrak{E} + 3\mathfrak{E}^2$, or $\Phi' + 3(H' - \frac{1}{3}I\Theta')^2$, viz. it is

$$IU'^2 - 3H'^2 + I^2\Theta'^2 + 12J\Theta'U' + 2I\Theta'H'$$

or, finally, it is

$$IU'^2 + 9H'^2 + \frac{1}{3}I^2\Theta'^2 + 12J\Theta'U' - 2I\Theta'H',$$

which is equal to

$$\begin{aligned} &IU'^2 + \frac{1}{3}I^2\Theta'^2 + 12J\Theta'U', \\ &= \frac{1}{I}[(IU' + 6J\Theta')^2 + \frac{1}{3}(I^3 - 27J^2)\Theta'^2]; \end{aligned}$$

so that the condition in order that this invariant may be equal to zero is

$$IU' + \left[6J \pm 2\sqrt{-\frac{1}{3}(I^3 - 27J^2)}\right]\Theta' = 0,$$

which agrees with a result of M. Hermite's.

There should, I think, be an identical equation of the form

$$JU'^3 - IU'^2H' + 4H'^3 + M\Theta' = -\Phi'^2,$$

which would serve to express the square of the invariant Φ' in terms of the other invariants U', H', Θ', I, J ; but assuming that such an equation exists, the form of the factor M remains to be ascertained. The invariant J (cubinvariant) of the form $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$ contains Φ'^2 , and it would be necessary to make use of the identical equation just referred to in order to reduce it to its simplest form; and (this being so) I have not sought for the expression of the cubinvariant of $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \mid y, 1)^4$.

For the quintic we have the equations

$$(a, b, c, d, e, f \mid x, 1)^5 = 0,$$

$$\begin{aligned} y &= (ax + b) B \\ &+ (ax^2 + 2bx + 3c) C \\ &+ (ax^3 + 3bx^2 + 6cx + 4d) D \\ &+ (ax^4 + 4bx^3 + 10cx^2 + 10dx + 5e) E, \end{aligned}$$

and this leads to the system of equations

$$\left| \begin{array}{cccccc} y - bB - 4cC - 6dD - 5eE & -aB - 5bC - 10cD - 10dE & -aC - 5bD - 10cE & -aD - 5bE & -aE \\ fE & y - bB - 4cC - 6dD + eE & -aB - 5bC - 10cD & -aC - 5bD & -aD \\ fD & 5eD + fE & y - bB - 4cC + 4dD + eE & -aB - 5bC & -aC \\ fC & 5eC + fD & 10dC + 5eD + fE & y - bB + 6cC + 4dD + eE & -aB \\ fB & 5eB + fC & 10dB + 5eC + fD & 10cB + 10dC + 5eD + fE & y + 4bB + 6cC + 4dD + eE \end{array} \right| = 0,$$

and the transformed equation is obtained by equating to zero the determinant formed out of the matrix contained in this equation.

The determinant in question was calculated by the formula

$$\square = \begin{array}{rcl} & \text{Div.} & \\ & -12 \cdot 345 & 1 \\ & +13 \cdot 245 & 2 \\ & -14 \cdot 235 & 3 \\ & +15 \cdot 234 & 4 \\ & -23 \cdot 145 & 5 \\ & +24 \cdot 135 & 6 \\ & -25 \cdot 134 & 7 \\ & -34 \cdot 125 & 8 \\ & +35 \cdot 124 & 9 \\ & -45 \cdot 123 & 10, \end{array}$$

where the duadic symbols refer to the first and fifth columns, viz. 12 is the determinant formed out of the lines 1 and 2 of these columns, and so for the other like symbols; and the triadic symbols refer to the second, third, and fourth columns, viz. 345 is the determinant formed out of the lines 3, 4, 5 of these columns, and so for the other like symbols.

The ten divisions were separately calculated. It is to be noticed that these divisions other than 4 and 6 correspond to each other in pairs, while each of the divisions 4 and 6 corresponds to itself, as thus:

$$\begin{array}{rcl} \text{Div. } 1, & -10 \\ 2, & -9 \\ 3, & -7 \\ 5, & -8 \\ 4, & -4 \\ 6, & -6, \end{array}$$

viz. if in the place of

$$y; \ a, b, c, d, e, f; \ B, C, D, E,$$

we write

$$-y; \ f, e, d, c, b, a; \ E, D, C, B,$$

then division 1 becomes division 10 with its sign reversed, and so for divisions 2 and 9, 3 and 7, 5 and 8; while each of the divisions 4 and 6 is unaltered, except that the sign is reversed. But the corresponding divisions were each of them calculated, and the property in question was used as a verification. Another very convenient verification, which was employed for the several divisions, was obtained by putting

$$a = b = c = d = e = f = B = C = D = E = 1,$$

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upon which supposition the determinant becomes

$$\begin{vmatrix} y-26, & -16, & -6, & -1 & \\ 1, & y-10, & -16, & -6, & -1 \\ 1, & 6, & y, & -6, & -1 \\ 1, & 6, & 16, & y+10, & -1 \\ 1, & 6, & 16, & 16, & y+15 \end{vmatrix},$$

and the values of the ten divisions respectively are

$y^3,$	$y^2,$	$y,$	1	Div.
26,	-544,	+4608,	-16384	1
96,	-540,	+6144,	-24576	2
36,	+672,	-3584,	+24576	+15 3
1,	0,	0,	0	4
26,	0,	0,	0	5
6,	0,	0,	0	6
16,	+544,	+3584,	+24576	7
+576,				8
+288,	+6144,	+16384		9
+4608,	+24576			10
1,	0,	0,	0,	0

[The exact integer entries in this table are reproduced from the scan with approximate fidelity only: the original printing on this page is broken in two places and the OCR is correspondingly unreliable; consult the IA scan, p. 384, for the authoritative listing. The sum of the ten divisions equals y^5 , in agreement with the verification $a = b = \cdots = E = 1$ stated below.]

A verification similar to this was in fact employed at each step of the calculation of a division: viz. in forming a product such as $(\lambda X + \mu Y + \&c.)(\lambda' X + \mu' Y + \&c.)$, where $\lambda, \mu, \&c., \lambda', \mu', \dots$ are numerical coefficients, and $X, Y, \&c.$ are monomial products of a, b, c, d, e, f and B, C, D, E , the sum of the numerical coefficients of the product is $(\lambda + \mu + \&c.)(\lambda' + \mu' + \&c.)$.

It was of course necessary to employ such verifications, as a test of the correctness of the several divisions, before proceeding to collect them together, but the collection itself affords an exceedingly good ultimate verification. The following is an exemplification: the terms in y which involve the product $BCDE$ are obtained by the collection of the corresponding terms in the ten divisions, as follows:

[Multi-column tabular collection by monomial in $(a, b, \dots, f)$; OCR illegible. Net coefficient of $BCDE$ verifies to +1228; cross-sum check yields 0.]

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where it may be remarked that the greater part, but not all, of the component coefficients are divisible by 5. I soon observed in the process of summing the ten divisions that all the resulting coefficients should be divisible by 5 (the only exception is as to the terms in y^4 which contain B^4, C^4, D^4 , and E^4

respectively), and the circumstance that they are so in each particular instance is as far as it goes a verification, which, however, only applies to those of the component coefficients which are not themselves divisible by 5. But it was known *à priori* (I will presently show how this is so) that the sum of the resulting coefficients should be equal to zero, and that they are so in fact is a verification as to all the coefficients. The foregoing specimen term $BCDE$ is one which remains unaltered when B, C, D, E are changed into E, D, C, B ; and on making the further change a, b, c, d, e, f into f, e, d, c, b, a , the coefficient of $BCDE$ remains, as it should do, unaltered; this is a verification of the coefficients of the terms $ace^2, b^2df; ad^2e, b^2cf$, which are respectively interchanged by the substitution in question, but not of the other terms $c^2f, abef, acdf, b^2e^2, bcdf$, which are respectively unaltered by the substitution. I did not employ what would have been another convenient verification of the several divisions, viz. the comparison of their values on putting therein $a = b = c = d = e = f = 1$, with the corresponding values as calculated independently from the determinant

$$\begin{vmatrix} y - B - 4C - 6D - 5E & -B - 5C - 10D - 10E & -C - 5D - 10E & -D - 5E & -E \\ E & y - B - 4C - 6D + E & -B - 5C - 10D & -C - 5D & -D \\ D & 5D + E & y - B - 4C + 4D + E & -B - 5C & -C \\ C & 5C + D & 10C + 5D + E & y - B + 6C + 4D + E & -B \\ B & 5B + C & 10B + 5C + D & 10B + 10C + 5D + E & y + 4B + 6C + 4D + E \end{vmatrix}$$

the calculation of the ten divisions of this determinant would however itself have been a work of some labour.

The last-mentioned determinant is $= y^5$; in fact, equating it to zero, we have the transformed equation corresponding to the system of equations

$$(1, 1, 1, 1, 1, 1 \mid x, 1)^5 = 0,$$

$$y = (x + 1)B + (x^2 + 5x + 4)C + (x^3 + 5x^2 + 10x + 6)D + (x^4 + 5x^3 + 10x^2 + 10x + 4)E.$$

But the first of these equations is $(x + 1)^5 = 0$, and the second is

$$y = (x + 1)[B + (x + 4)C + (x^2 + 4x + 6)D + (x^3 + 4x^2 + 6x + 4)E],$$

so that for each of the five equal roots $x = -1$, we have $y = 0$, or the transformed equation in y is $y^5 = 0$.

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And since upon writing $a = b = c = d = e = f = 1$ the transformed equation becomes $y^5 = 0$, it is clear that in the coefficient of any monomial product of B, C, D, E , the sum of the numerical coefficients of the several monomial products of a, b, c, d, e, f must be $= 0$, which is the property above referred to as affording a verification of the calculated expression of the transformed equation.

The final result is that the equations

$$(a, b, c, d, e, f \mid x, 1)^5 = 0,$$

$$\begin{aligned} y &= (ax + b) B \\ &+ (ax^2 + 2bx + 3c) C \\ &+ (ax^3 + 3bx^2 + 6cx + 4d) D \\ &+ (ax^4 + 4bx^3 + 10cx^2 + 10dx + 5e) E \end{aligned}$$

give for the transformed equation in y

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0,$$

where $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ are functions of the coefficients (a, b, c, d, e, f) and (B, C, D, E) of the degrees 2, 3, 4, 5 respectively in the latter, and each homogeneous of degree 2, 3, 4, 5 in the former. The full developed expressions are displayed on pages 386–390 of the original as four multi-column tables of monomial coefficients, viz. one for each of $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$, each table indexed in one direction by the monomials of (B, C, D, E)

of the appropriate weight, and in the other direction by the monomials of (a, b, c, d, e, f) . We reproduce these tables in the form of placeholders, since the OCR of these dense tabular arrays is not faithful and the surrounding text suffices to identify their content:

[Table for $\frac{1}{4}\mathfrak{C}$ (degree 2 in B, C, D, E): header monomials $B^2, BC, BD, C^2, BE, CD, CE, DE, D^2, E^2$; row monomials of degree 2 in a, b, c, d, e, f ($ac, b^2, ad, bc, \dots, e^2$ etc.). Coefficients are small integers ranging over $\{-16, \dots, +12\}$.]

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[Continuation of the table for $\frac{1}{4}\mathfrak{C}$, and beginning of the table for $\frac{1}{8}\mathfrak{D}$ (degree 3 in B, C, D, E): header monomials $B^3, B^2C, BC^2, C^3, B^2D, BCD, B^2E, \dots, DE^2, E^3$; row monomials of degree 3 in a, b, c, d, e, f . Coefficients are small integers, generally divisible by 5 except in the four leading positions B^3, C^3, D^3, E^3 .]

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[Table for $\frac{1}{16}\mathfrak{E}$ (degree 4 in B, C, D, E): header monomials $B^4, B^3C, B^3D, B^2C^2, B^3E, B^2CD, \dots, DE^3, E^4$; row monomials of degree 4 in a, b, c, d, e, f ($a^2ce, a^2d^2, a^2cf, \dots$). Coefficients are integers, generally larger in magnitude than for $\mathfrak{C}$ or $\mathfrak{D}$, e.g. $\pm 70, \pm 195, \pm 1000$.]

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[Continuation of the table for $\frac{1}{16}\mathfrak{E}$, and beginning of the table for $\frac{1}{32}\mathfrak{F}$ (degree 5 in B, C, D, E): header monomials $B^5, B^4C, B^4D, B^3C^2, B^4E, B^3CD, \dots, DE^4, E^5$; row monomials of degree 5 in a, b, c, d, e, f . Coefficients are larger still, e.g. entries with values such as $\pm 245, \pm 5070, \pm 23010, \pm 25475$ appear in the column-sum rows.]

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[Conclusion of the table for $\frac{1}{32}\mathfrak{F}$, with the final column sums that verify (after multiplication by $\frac{1}{5}$) the divisibility property stated above.]

Upon writing $(B, C, D, E) = (x^4, xy^3, xy^2, y^4)$, the foregoing values of $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ become covariants of the quintic $(a, b, c, d, e, f \mid x, y)^5$. In fact [using for the covariants of the quintic the notation of §§ 141 and 143, A the quintic itself,

$$\begin{aligned}\frac{1}{4}\mathfrak{C} &= 2C, \\ \frac{1}{8}\mathfrak{D} &= 2F, \\ \frac{1}{16}\mathfrak{E} &= A^2E - 2C^2, \\ \frac{1}{32}\mathfrak{F} &= A^2F - 2CF.\end{aligned}$$

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This [the circumstance that the values thus become covariants of the quintic] is a further verification of the foregoing result.

I will conclude by showing how the formula may be applied to the reduction of the general quintic equation to Mr Jerrard's form $x^5 + ax + b = 0$. It was long ago remarked by Professor Sylvester that Tschirnhausen's Transformation, in its original form, gave the means of effecting this reduction. In fact, if the transforming equation be

$$y = \alpha + \beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4,$$

then the equation in y is of the form

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0,$$

where $\mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ are, in regard to $\alpha, \beta, \gamma, \delta, \varepsilon$, of the degrees 1, 2, 3, 4, 5 respectively. And by assuming, say α a linear function of $\beta, \gamma, \delta, \varepsilon$, we may make $\mathfrak{B} = 0$, and we have then $\mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}$ functions of the degrees 2, 3, 4, 5 respectively of the quantities $\beta, \gamma, \delta, \varepsilon$: and these can be determined by means of a quadric

equation and a cubic equation in such manner that $\mathfrak{C} = 0, \mathfrak{D} = 0$, in which case the equation in y will be of the required form. For considering $\beta, \gamma, \delta, \varepsilon$ as the coordinates of a point in space, the equations $\mathfrak{C} = 0, \mathfrak{D} = 0$ will be the equations of a quadric surface and a cubic surface respectively, and if $\beta, \gamma, \delta, \varepsilon$ be the coordinates of a point on the curve of intersection, the required conditions will be satisfied. And by combining with the equation of the quadric surface, the equation of any tangent plane thereto (or by the different process which is made use of in the sequel), we may, by means of a quadric equation, find a generating line of the quadric surface, and then, by means of a cubic equation, a point of intersection of this line with the cubic surface, i.e. a point the coordinates whereof give the required values $\beta, \gamma, \delta, \varepsilon$. And similarly for the new form of Tschirnhausen's Transformation; the only difference being that, starting with an equation in y which contains the four arbitrary quantities B, C, D, E , we obtain in the first instance an equation

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \mid y, 1)^5 = 0$$

wanting the second term. And then B, C, D, E are to be so determined that $\mathfrak{C} = 0, \mathfrak{D} = 0$.

To proceed with the reduction, I write the foregoing value of $\mathfrak{C}$ in the form

$$\frac{1}{4}\mathfrak{C} = \left(\begin{array}{cccc} 4ac - 4b^2 & 6ad - 6bc & 4ae - 4bd & af - be \\ 6ad - 6bc & 4ae + 20bd - 24c^2 & af + 15be - 16cd & 4bf - 4ce \\ 4ae - 4bd & af + 15be - 16cd & 4bf + 20ce - 24d^2 & 6cf - 6de \\ af - be & 4bf - 4ce & 6cf - 6de & 4df - 4e^2 \end{array} \middle| B, C, D, E \right)^2,$$

which for shortness may be represented by

$$\frac{1}{4}\mathfrak{C} = \left(\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & p \end{array} \middle| B, C, D, E \right)^2,$$

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or say

$$\frac{1}{4}\mathfrak{C} = (\Omega \mid B, C, D, E)^2.$$

Now, by a formula in my memoir "On the Automorphic Linear Transformation of a Bipartite Quadric Function," *Phil. Trans.*, vol. CXLVIII. (1858), see p. 44, [153], if T denote any skew symmetric matrix of the order 4, then if

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)(\Omega + T)^{-1}\Omega \mid B', C', D', E'),$$

in which formula $\Omega^{-1}, \Omega - T, (\Omega + T)^{-1}, \Omega$ are all matrices which are to be compounded together into a single matrix, we have identically

$$(\Omega \mid B, C, D, E)^2 = (\Omega \mid B', C', D', E')^2.$$

Let Q denote the determinant $|\Omega + T|$, then the terms of the matrix $(\Omega + T)^{-1}$ are respectively divided by Q , and we may write

$$(\Omega + T)^{-1} = \frac{1}{Q} Q(\Omega + T)^{-1},$$

where $Q(\Omega + T)^{-1}$ is the matrix obtained from the matrix $(\Omega + T)^{-1}$ by multiplying each term by Q , the terms of $Q(\Omega + T)^{-1}$ being thus rational and integral functions of the terms of the matrix $(\Omega + T)$. Hence if, instead of the before-mentioned relation between (B, C, D, E) and (B', C', D', E') , we assume

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', D', E'),$$

we find

$$(\Omega \mid B, C, D, E)^2 = Q^2(\Omega \mid B', C', D', E')^2.$$

If here the matrix T is such that we have $Q = 0$, i.e. $\text{Det.}(\Omega + T) = 0$ (which is a quadric relation between the terms of the skew matrix, that is, each term is contained therein in the first and second powers only), then the equation becomes

$$(\Omega \mid B, C, D, E)^2 = 0.$$

It is clear that this can only be the case in consequence of the coefficients of trans- formation in the equation

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', D', E')$$

being such that there shall exist at least two linear relations between the quantities (B, C, D, E) , and I assume (without stopping to prove it) that they are such that the number of such linear relations is in fact two. That is, the last-mentioned equation establishes between the quantities (B, C, D, E) two linear relations, in virtue whereof $\mathfrak{C} = 0$. And this being so, we may, without loss of generality, write $D' = 0$, $E' = 0$, or put

$$(B, C, D, E) = (\Omega^{-1}(\Omega - T)Q(\Omega + T)^{-1}\Omega \mid B', C', 0, 0);$$

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so that B, C, D, E are linear functions of B', C' , such that $\mathfrak{C} = 0$. And then substituting these values for (B, C, D, E) , we find $\mathfrak{D}$ a cubic function of B', C' ; so that, putting $\mathfrak{D} = 0$, we have a cubic equation to determine the ratio $B' : C'$.

The foregoing reasoning presents no real difficulty, but it is expressed by means of a very condensed notation, and it may be proper to illustrate it by the case of the quadric function $x^2 + y^2 + z^2$. Considering the equations

$$\begin{aligned} x &= (1 + \lambda^2 - \mu^2 - \nu^2)x' + 2(\lambda\mu - \nu)y' + 2(\lambda\nu + \mu)z', \\ y &= 2(\lambda\mu + \nu)x' + (1 - \lambda^2 + \mu^2 - \nu^2)y' + 2(\mu\nu - \lambda)z', \\ z &= 2(\nu\lambda - \mu)x' + 2(\mu\nu + \lambda)y' + (1 - \lambda^2 - \mu^2 + \nu^2)z', \end{aligned}$$

these equations, if the expressions for x, y, z had been divided by $1 + \lambda^2 + \mu^2 + \nu^2$, would have given

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2.$$

Hence they actually do give

$$x^2 + y^2 + z^2 = (1 + \lambda^2 + \mu^2 + \nu^2)^2(x'^2 + y'^2 + z'^2);$$

or if

$$1 + \lambda^2 + \mu^2 + \nu^2 = 0,$$

they give

$$x^2 + y^2 + z^2 = 0.$$

But if

$$1 + \lambda^2 + \mu^2 + \nu^2 = 0,$$

then

$$\begin{vmatrix} 1 + \lambda^2 - \mu^2 - \nu^2 & 2(\lambda\mu - \nu) & 2(\lambda\nu + \mu) \\ 2(\lambda\mu + \nu) & 1 - \lambda^2 + \mu^2 - \nu^2 & 2(\mu\nu - \lambda) \\ 2(\nu\lambda - \mu) & 2(\mu\nu + \lambda) & 1 - \lambda^2 - \mu^2 + \nu^2 \end{vmatrix} = 0,$$

so that we have

$$x : y : z = 1 + \lambda^2 - \mu^2 - \nu^2 : 2(\lambda\mu + \nu) : 2(\nu\lambda - \mu),$$

which is the same result as would have been found by writing $y' = z' = 0$, and which comes to saying that x, y, z are not independent, but are connected by two linear relations.

The equation $\text{Det.}(\Omega + T) = 0$, written at length, will be

$$\begin{vmatrix} a, & h - \tau, & g + \sigma, & l + \lambda \\ h + \tau, & b, & f - \rho, & m + \mu \\ g - \sigma, & f + \rho, & c, & n + \nu \\ l - \lambda, & m - \mu, & n - \nu, & p \end{vmatrix} = 0,$$

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where $\lambda, \mu, \nu, \rho, \sigma, \tau$ are the arbitrary constituents of the skew matrix; or developing, this is

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & p \end{vmatrix}$$

terms quadratic in $\lambda, \mu, \nu, \rho, \sigma, \tau$ with coefficients
+ $\left(\begin{array}{l} \text{bilinear in the entries } a, b, c, f, g, h, l, m, n, p \text{ of } \Omega, \\ \text{viz. the developed expansion as displayed in the original.} \end{array} \right) + (\lambda\rho + \mu\sigma + \nu\tau)^2 = 0,$

the first term whereof, substituting for $(a, b, c, f, g, h, l, m, n, p)$ their values, is in fact equal to the discriminant $a^4f^4 + \&c.$ of the quintic $(a, b, c, d, e, f \mid x, y)^5$. There is no loss of generality in putting all but two of the quantities $(\lambda, \mu, \nu, \rho, \sigma, \tau)$ equal to zero; in fact this leaves in the formulae a single arbitrary quantity, which is the right number, since the ratios $B : C : D : E$ have to satisfy only the two conditions $\mathfrak{C} = 0, \mathfrak{D} = 0$.

[An Addition of a half page, dated Nov. 10, 1862, and referring to the Memoir “On a New Auxiliary Equation in the Theory of Equations of the Fifth Order”, 268, is printed at the end of that paper, ante p. 324.]

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ON THE ANALYTICAL THEORY OF THE CONIC.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLII. (for the year 1862), pp. 639–662.
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THE decomposition into its linear factors of a decomposable quadric function cannot be effected in a symmetrical manner otherwise than by formulae containing super- numerary arbitrary quantities; thus, for a binary quadric (which of course is always decomposable) we have

$$(a, b, c \mid x, y)^2 = \frac{1}{(a, b, c \mid x', y')^2} \{ (a, b, c \mid x, y \mid x', y') \pm \sqrt{ac - b^2} (xy' - x'y) \}^2$$

or the expression for a linear factor is

$$\frac{1}{\sqrt{(a,b,c|x',y')^2}} \{ (a,b,c | x,y | x',y') \pm \sqrt{ac-b^2} (xy' - x'y) \},$$

which involves the arbitrary quantities (x', y') . And this appears to be the reason why, in the analytical theory of the conic, the questions which involve the decomposition of a decomposable ternary quadric have been little or scarcely at all considered: thus, for instance, the expressions for the coordinates of the points of intersection of a conic by a line (or say the line-equations of the two ineunts), and the equations for the tangents (separate each from the other) drawn from a given point not on the conic, do not appear to have been obtained. These questions depend on the decomposition of a decomposable ternary quadric, which decomposition itself depends on that for the simplest case, when the quadric is a perfect square. Or we may say that in the first instance they depend on the transformation of a given quadric function $U = (* | x, y, z)^2$ into the form $W^2 + V$, where W is a linear function, given save as to a constant factor (that is, $W = 0$ is the equation of a given line), and V is a decomposable quadric function, which is ultimately decomposed into its linear factors, $= QR$, so that we have $U = W^2 + QR$.

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The formula for this purpose, which is exhibited in the eight different forms I, II, III, IV, I (bis), II (bis), III (bis), IV (bis), is the analytical basis of the whole theory; and the greater part of the memoir relates to the establishment of these forms.

The solution of the geometrical questions above referred to is (as shown in the memoir) involved in and given immediately by these forms. It is also shown that the formulae are greatly simplified in the case e.g. of tangents drawn to a conic from a point in a conic having double contact with the first-mentioned conic, and that in this case they lead to the linear Automorphic Transformation of the ternary quadric. The memoir concludes with some formulae relating to the case of two conics, which however is treated of in only a cursory manner.

Article Nos. 1–17, relating to a single conic.

1. The point-equation of the conic is

$$(a, b, c, f, g, h | x, y, z)^2 = 0,$$

which expresses that the point (x, y, z) is an ineunt of the conic.

The line-equation of the same conic is

$$\begin{vmatrix} \xi, \eta, \zeta, 0 \\ \xi, a, h, g \\ \eta, h, b, f \\ \zeta, g, f, c \end{vmatrix} = 0,$$

or putting

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

the line-equation is

$$(A, B, C, F, G, H | \xi, \eta, \zeta)^2 = 0,$$

which expresses that the line (ξ, η, ζ) (that is, the line the point-equation whereof is $\xi x + \eta y + \zeta z = 0$) is a tangent of the conic. We are thus in the analytical theory of the conic concerned with the quadrics $(a, b, c, f, g, h | x, y, z)^2$ and $(A, B, C, F, G, H | \xi, \eta, \zeta)^2$, which are the characteristics or nilfactums of these equations respectively.

2. I put also

$$K = \begin{vmatrix} a, h, g \\ h, b, f \\ g, f, c \end{vmatrix},$$

or, what is the same thing,

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

3. It may be convenient to notice that when $(a, \dots | x, y, z)^2$ breaks up into factors, the conic the equation whereof is $(a, \dots | x, y, z)^2 = 0$ becomes a pair of lines; and that, when $(a, \dots | x, y, z)^2$ is a perfect square, the conic becomes a pair of coincident lines, or say a twofold line. But a pair of lines, distinct or coincident, cannot be represented by a line-equation. The analytical formulae presently given show that in the former case $(A, \dots | \xi, \eta, \zeta)^2$ is the square of a linear function, which equated to zero gives the line-equation of the point of intersection of the two lines, or node of the conic; and the equation $(A, \dots | \xi, \eta, \zeta)^2 = 0$ accordingly represents such point considered as a pair of coincident points, or say a twofold point. But in the latter case, where the conic is a twofold line, $(A, \dots | \xi, \eta, \zeta)^2$ is identically equal to zero, and the line-equation $(A, \dots | \xi, \eta, \zeta)^2 = 0$ is a mere identity $0 = 0$, thus ceasing to have any signification at all. And the like remarks apply to the conic as represented by the line-equation $(A, \dots | \xi, \eta, \zeta)^2 = 0$, the conic here breaking up into a pair of distinct or coincident points, &c.

4. It is proper to remark also that

$$(a, \dots | x', y', z' | x, y, z) = 0$$

is the equation of the polar of the point (x', y', z') in regard to the conic, and that

$$(A, \dots | \xi', \eta', \zeta' | \xi, \eta, \zeta) = 0$$

is the line-equation of the pole of the line (ξ', η', ζ') ; or, what is the same thing, the point-coordinates of the pole are

$$A\xi' + H\eta' + G\zeta' : H\xi' + B\eta' + F\zeta' : G\xi' + F\eta' + C\zeta'.$$

5. The inverse matrix is

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix}^{-1} = \frac{1}{K} \begin{pmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{pmatrix},$$

but it is convenient to disregard the factor $\frac{1}{K}$, and speak of (A, B, C, F, G, H) as the inverse or reciprocal coefficients. The equation just written down implies the relations $Aa + Hh + Gg = K$, $Ah + Hb + Gf = 0$, &c., which may be arranged in two different ways as a system of nine equations.

6. We have also

$$(BC - F^2, CA - G^2, AB - H^2, GH - AF, HF - BG, FG - CH) = K(a, b, c, f, g, h),$$

and

$$ABC - AF^2 - BG^2 - CH^2 + 2FGH = K^2,$$

which are well-known theorems.

7. I notice also the theorem

$$(a, \dots | x, y, z)^2 \cdot (a, \dots | x', y', z')^2 - [(a, \dots | x, y, z | x', y', z')]^2 = (A, \dots | yz' - y'z, zx' - z'x, xy' - x'y)^2,$$

which is much used in the sequel: it may be mentioned, in passing, that this is included in the more general theorem

$$\begin{vmatrix} (a, \dots | x, y, z | l, m, n) & (a, \dots | x', y', z' | l, m, n) \\ (a, \dots | x, y, z | l', m', n') & (a, \dots | x', y', z' | l', m', n') \end{vmatrix} \\ = (A, \dots | yz' - y'z, zx' - z'x, xy' - x'y | mn' - m'n, nl' - n'l, lm' - l'm),$$

which is at once deducible from

$$\begin{vmatrix} Ll + Mm + Nn & L'l + M'm + N'n \\ Ll' + Mm' + Nn' & L'l' + M'm' + N'n' \end{vmatrix} = (MN' - M'N)(mn' - m'n) + (NL' - N'L)(nl' - n'l) + (LM' - L'M)(lm' - l'm),$$

by writing therein

$$(L, M, N) = (ax + hy + gz, hx + by + fz, gx + fy + cz),$$

$$(L', M', N') = (ax' + hy' + gz', hx' + by' + fz', gx' + fy' + cz').$$

8. Suppose now that

$$(a, b, c, f, g, h \mid x, y, z)^2$$

breaks up into factors, or say that we have

$$(a, b, c, f, g, h \mid x, y, z)^2 = 2(\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z),$$

the values of the coefficients $(a, \dots)$ then are

$$(a, b, c, f, g, h) = (2\alpha\alpha', 2\beta\beta', 2\gamma\gamma', \beta\gamma' + \beta'\gamma, \gamma\alpha' + \gamma'\alpha, \alpha\beta' + \alpha'\beta),$$

and forming from these the inverse coefficients $(A, \dots)$ and the discriminant K , we find

$$(A, B, C, F, G, H) = -(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta)^2,$$

$$K = 0.$$

9. The last-mentioned equation, $K = 0$, is the condition in order that $(a, \dots \mid x, y, z)^2$ may break up into factors; and when it does so, we have

$$(A, \dots \mid \xi, \eta, \zeta)^2 = -[(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta \mid \xi, \eta, \zeta)]^2,$$

that is, $(a, \dots \mid x, y, z)^2$ breaking up into factors, $(A, \dots \mid \xi, \eta, \zeta)^2$ is a perfect square; and equating it to zero, we have

$$[(\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta \mid \xi, \eta, \zeta)]^2 = 0;$$

which, (ξ, η, ζ) being line-coordinates, gives (as a two-fold point) the point of intersection of the lines $(\alpha, \beta, \gamma), (\alpha', \beta', \gamma')$, that is, the lines $\alpha x + \beta y + \gamma z = 0, \alpha' x + \beta' y + \gamma' z = 0$.

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10. If $(a, \dots \mid x, y, z)^2$ is a perfect square, then $\alpha' : \beta' : \gamma' = \alpha : \beta : \gamma$; whence not only, as before, $K = 0$, but the coefficients (A, B, C, F, G, H) all vanish (this implies the first-mentioned condition, $K = 0$); and the line-equation $(A, \dots \mid \xi, \eta, \zeta)^2 = 0$ becomes the mere identity $0 = 0$.

11. Conversely if $K = 0$, then $(a, \dots \mid x, y, z)^2$ breaks up into factors; and if (A, B, C, F, G, H) all vanish, then $(a, \dots \mid x, y, z)^2$ is a perfect square. The conclusions stated *ante*, No. 3, are thus sustained.

12. I assume, first, that $(a, \dots \mid x, y, z)^2$ is a perfect square (No. 13); and secondly, that it breaks up into factors (No. 14); and I proceed to inquire how in the one case the root, and in the other case the factors, can be determined in a symmetrical form.

13. Considering the before-mentioned identical equation

$$(a, \dots \mid x, y, z)^2 \cdot (a, \dots \mid x', y', z')^2 - [(a, \dots \mid x, y, z \mid x', y', z')]^2 = (A, \dots \mid yz' - y'z, zx' - z'x, xy' - x'y)^2,$$

if $(a, \dots \mid x, y, z)^2$ is a perfect square, then by what precedes, the right-hand side of the equation vanishes, and we have

$$(a, \dots \mid x, y, z)^2 = \frac{[(a, \dots \mid x, y, z \mid x', y', z')]^2}{(a, \dots \mid x', y', z')^2},$$

and the root of $(a, \dots \mid x, y, z)^2$ is thus seen to be

$$\pm \frac{(a, \dots \mid x, y, z \mid x', y', z')}{\sqrt{(a, \dots \mid x', y', z')^2}},$$

an expression which involves the quantities (x', y', z') , the values whereof may be assumed at pleasure without altering the value of the expression. For instance, assuming for (x', y', z') the values $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ successively, the different values of the expression are

$$\frac{ax + hy + gz}{\sqrt{a}}, \quad \frac{hx + by + fz}{\sqrt{b}}, \quad \frac{gx + fy + cz}{\sqrt{c}}.$$

But if, as assumed, $(a, \dots | x, y, z)^2$ be a perfect square $= (\alpha x + \beta y + \gamma z)^2$, then

$$(a, b, c, f, g, h) = (\alpha^2, \beta^2, \gamma^2, \beta\gamma, \gamma\alpha, \alpha\beta),$$

and each of the foregoing values becomes equal to the root $\alpha x + \beta y + \gamma z$. It is somewhat singular that it is not possible to obtain symmetrical formulae without employing in this manner supernumerary arbitrary quantities such as (x', y', z') .

14. Next, if $(a, \dots | x, y, z)^2$, instead of being a perfect square, only breaks up into factors, then in the foregoing identical equation the right-hand side is a perfect square, and by the formula just obtained its value is

$$\frac{[(A, \dots | yz' - y'z, zx' - z'x, xy' - x'y | X, Y, Z)]^2}{(A, \dots | X, Y, Z)^2}$$

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where (X, Y, Z) are supernumerary arbitrary quantities. The identical equation then gives

$$(a, \dots | x, y, z)^2 = \frac{1}{(a, \dots | x', y', z')^2} \left[[(a, \dots | x, y, z | x', y', z')]^2 + \frac{[(A, \dots | yz' - y'z, \dots | X, Y, Z)]^2}{(A, \dots | X, Y, Z)^2} \right]$$

and consequently

$$(a, \dots | x, y, z)^2 = \frac{1}{(a, \dots | x', y', z')^2} \text{Product of} \\ \left\{ (a, \dots | x, y, z | x', y', z') \pm \frac{(A, \dots | X, Y, Z | yz' - y'z, zx' - z'x, xy' - x'y)}{\sqrt{(A, \dots | X, Y, Z)^2}} \right\},$$

a formula which exhibits the decomposition of $(a, \dots | x, y, z)^2$ assumed to be a function which breaks up into factors; the formula contains the two sets of supernumerary arbitrary quantities (x', y', z') and (X, Y, Z) . It will be remembered that $(A, \dots)$ denotes the system of inverse or reciprocal coefficients $(bc - f^2, \dots)$.

15. Consider the formula

$$(a, b, c, f, g, h | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2 = (a, b, c, f, g, h | \xi', \eta', \zeta')^2,$$

which gives

$$\begin{aligned} a &= c\eta^2 + b\zeta^2 - 2f\eta\zeta, \\ b &= a\zeta^2 + c\xi^2 - 2g\zeta\xi, \\ c &= b\xi^2 + a\eta^2 - 2h\xi\eta, \\ f &= -a\eta\zeta - f\xi^2 + g\xi\eta + h\zeta\xi, \end{aligned}$$

and from these we deduce

$$\begin{aligned} g &= -b\zeta\xi + f\xi\eta - g\eta^2 + h\eta\zeta, \\ h &= -c\xi\eta + f\zeta\xi + g\eta\zeta - h\zeta^2; \end{aligned}$$

viz.

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix} \begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

viz. $a\xi + h\eta + g\zeta = 0$, &c.

Also

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch) = (\xi', \eta', \zeta')^2 (A, B, C, F, G, H | \xi, \eta, \zeta)^2,$$

that is

$$bc - f^2 = \xi^2(A, B, C, F, G, H | \xi, \eta, \zeta)^2, \quad \&c.$$

Hence also

$$(bc - f^2, \dots | l, m, n)^2 = (l\xi + m\eta + n\zeta)^2 (A, \dots | \xi, \eta, \zeta)^2,$$

$$(bc - f^2, \dots | l, m, n | l', m', n') = (l\xi + m\eta + n\zeta)(l'\xi + m'\eta + n'\zeta) (A, \dots | \xi, \eta, \zeta)^2;$$

and moreover

$$abc - af^2 - bg^2 - ch^2 + 2fgh = 0.$$

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16. The last equation shows that $(a, \dots | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2$, considered as a function of (ξ', η', ζ') , breaks up into factors. Or since the expression is not altered by interchanging (ξ', η', ζ') and (ξ, η, ζ) , the same expression, considered as a function of (ξ, η, ζ) , breaks up into factors. It is in fact easy to see that any quantic whatever, $(* | \eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2$, considered as a function of (ξ, η, ζ) , breaks up into linear factors; for in virtue of the equation $\xi'(\eta\zeta' - \eta'\zeta) + \eta'(\zeta\xi' - \zeta'\xi) + \zeta'(\xi\eta' - \xi'\eta) = 0$, any one of the quantities $\eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta$ can be expressed as a linear function of the other two; so that the quantic can be expressed as a linear function of any two of the three quantities; and *quel* homogeneous function of two quantities, it of course breaks up into factors, linear functions of these two quantities.

We may in all the formulae interchange (x', y', z') and (x, y, z) , writing (a', b', c', f', g', h') in the place of (a, b, c, f, g, h) .

17. Putting, in like manner,

$$(A, B, C, F, G, H | yz' - y'z, zx' - z'x, xy' - x'y)^2 = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} | x, y, z)^2,$$

we obtain

$$\begin{aligned}\mathfrak{A} &= Cy^2 + Bz^2 - 2Fyz, \\ \mathfrak{B} &= Az^2 + Cx^2 - 2Gzx, \\ \mathfrak{C} &= Bx^2 + Ay^2 - 2Hxy, \\ \mathfrak{F} &= -Ayz - Fx^2 + Gxy + Hzx, \\ \mathfrak{G} &= -Bzx + Fxy - Gy^2 + Hyz, \\ \mathfrak{H} &= -Cxy + Fzx + Gyz - Hz^2,\end{aligned}$$

viz.

$$\begin{pmatrix} \mathfrak{A}, & \mathfrak{H}, & \mathfrak{G} \\ \mathfrak{H}, & \mathfrak{B}, & \mathfrak{F} \\ \mathfrak{G}, & \mathfrak{F}, & \mathfrak{C} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

viz. $\mathfrak{A}x + \mathfrak{H}y + \mathfrak{G}z = 0$, &c.

Also

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \mathfrak{C}\mathfrak{A} - \mathfrak{G}^2, \mathfrak{A}\mathfrak{B} - \mathfrak{H}^2, \mathfrak{G}\mathfrak{H} - \mathfrak{A}\mathfrak{F}, \mathfrak{H}\mathfrak{F} - \mathfrak{B}\mathfrak{G}, \mathfrak{F}\mathfrak{G} - \mathfrak{C}\mathfrak{H}) = (x, y, z)^2 K(a, b, c, f, g, h | x, y, z)^2;$$

that is

$$\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2 = x^2 K(a, b, c, f, g, h | x, y, z)^2, \quad \&c.;$$

whence also

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \dots | \lambda, \mu, \nu)^2 = (\lambda x + \mu y + \nu z)^2 K(a, \dots | x, y, z)^2,$$

$$(\mathfrak{B}\mathfrak{C} - \mathfrak{F}^2, \dots | \lambda, \mu, \nu | \lambda', \mu', \nu') = (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) K(a, \dots | x, y, z)^2;$$

and moreover

$$\mathfrak{A}\mathfrak{B}\mathfrak{C} - \mathfrak{A}\mathfrak{F}^2 - \mathfrak{B}\mathfrak{G}^2 - \mathfrak{C}\mathfrak{H}^2 + 2\mathfrak{F}\mathfrak{G}\mathfrak{H} = 0.$$

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The last equation shows that $(A, \dots | yz' - y'z, zx' - z'x, xy' - x'y)^2$, considered as a function of (x', y', z') , breaks up into factors, or (what is the same thing) this expression, considered as a function of (x, y, z) , breaks up into factors; we may in all the formulae interchange (x, y, z) and (x', y', z') , writing $(\mathfrak{A}', \mathfrak{B}', \mathfrak{C}', \mathfrak{F}', \mathfrak{G}', \mathfrak{H}')$ in the place of $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$.

Article Nos. 18 to 28, relating to a single conic in connexion with a point or line.

18. I apply the decomposition formula to the function $(A, \dots | yz' - y'z, \dots)^2$, which, considered as a function of (x, y, z) , breaks up into factors. We have

$$(A, \dots | yz' - y'z, \dots)^2 = (\mathfrak{A}', \dots | x, y, z)^2 \\ = \frac{1}{(\mathfrak{A}', \dots | x, y, z)^2 \cdot x'^2 - y'^2 \cdot z'^2 (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2}.$$

But we have

$$(\mathfrak{A}', \dots | l, m, n)^2 = (A, \dots | mz' - ny', \dots)^2, \\ (\mathfrak{A}', \dots | l, m, n | x, y, z) = (A, \dots | mz' - ny', \dots | yz' - y'z, \dots), \\ (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | mz - ny, \dots | \lambda, \mu, \nu) = [x'(mz - ny) + y'(nx - lz) + z'(ly - mx)](\lambda x' + \mu y' + \nu z') K(a, \dots | x', y', z')^2, \\ (\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2 = (\lambda x' + \mu y' + \nu z')^2 K(a, \dots | x', y', z')^2,$$

and thence

$$\frac{(\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | mz - ny, \dots | \lambda, \mu, \nu)}{\sqrt{(\mathfrak{B}'\mathfrak{C}' - \mathfrak{F}'^2, \dots | \lambda, \mu, \nu)^2}} = \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2};$$

whence we have

$$(A, \dots | yz' - y'z, \dots)^2 = \frac{1}{(\mathfrak{A}', \dots | x, y, z)^2} \text{Product of} \\ \left\{ (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2} \right\};$$

and the identical equation

$$(a, \dots | x, y, z)^2 \cdot (a, \dots | x', y', z')^2 - [(a, \dots | x, y, z | x', y', z')]^2 = (A, \dots | yz' - y'z, \dots)^2$$

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now gives

$$(a, \dots | x, y, z)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of} \\ \left\{ \begin{array}{l} [(a, \dots | x, y, z | x', y', z')]^2 \\ \text{I.} \quad + \text{Quotient by } (A, \dots | mz' - ny', \dots)^2 \text{ of Product} \\ \left\{ (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix} \sqrt{K(a, \dots | x', y', z')^2} \right\} \end{array} \right.$$

where the Product part may also be written

$$(a, \dots | l, m, n | x', y', z') \cdot (a, \dots | x, y, z | x', y', z') \\ - (a, \dots | x', y', z')^2 \cdot (a, \dots | x, y, z | l, m, n) \\ \pm \sqrt{K(a, \dots | x', y', z')^2} \begin{vmatrix} x' & y' & z' \\ x & y & z \\ l & m & n \end{vmatrix}.$$

19. Writing in the formula I.

$$(a, h, g, f, h, b, c, f, g | x', y', z') = (\xi', \eta', \zeta'),$$

we have

$$\begin{pmatrix} a, h, g \\ h, b, f \\ g, f, c \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \xi' \\ \eta' \\ \zeta' \end{pmatrix}, \quad \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} \begin{pmatrix} \xi' \\ \eta' \\ \zeta' \end{pmatrix} = K \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix},$$

and thence

$$K(a, \dots | x', y', z')^2 = (A, \dots | \xi', \eta', \zeta')^2.$$

Assume

$$(a, \dots | x, y, z | x', y', z') = \xi'x + \eta'y + \zeta'z.$$

$$(l, m, n) = (\nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta');$$

then from the foregoing values of (x', y', z')

$$\begin{aligned} mz' - ny' &= \frac{1}{K} \left\{ (\lambda\zeta' - \nu\xi')(G\xi' + F\eta' + C\zeta') - (\mu\xi' - \lambda\eta')(H\xi' + B\eta' + F\zeta') \right\} \\ &= \frac{1}{K} \left\{ \lambda[\xi'(A\xi' + H\eta' + G\zeta') + \eta'(H\xi' + B\eta' + F\zeta') + \zeta'(G\xi' + F\eta' + C\zeta')] \right. \\ &\quad \left. - \lambda\xi'(A\xi' + H\eta' + G\zeta') - \mu\xi'(H\xi' + B\eta' + F\zeta') - \nu\xi'(G\xi' + F\eta' + C\zeta') \right\}. \end{aligned}$$

[continued p. 404]